1) Introduction: Longitudinal data

- repeated measurements on some subject at different time points (dense time points => functional data)
- Cohort Studies/Panel Studies: cohort of people with same characteristics and follow over time
- randomized trial: subjects randomly assigned to treatment groups

Terminology

- n_i, $i=1,\dots,N$: obs for subject i
- $t_{i1},\dots,t_{in_i}$: obs times for subject i
- balanced data: $n_i = n \quad \forall i$
- equally spaced: $t_{j+1} - t_j = d \quad \forall j$
- time-invariant Covariate: doesn't change, e.g. gender
- prosective measures: follow subject over time
- 2 sources of variation:
  - inter-subject (between subjects)
  - intra-subject (same subject over different measurements)
- longitudinal data more informative than cross-sectional, bc we see individual changes over time
  - ↳ better protection against confounders: time-constant confounders eliminated by taking differences

$$E(Y_{ij}) = \underbrace{\beta_0 + \beta_1 z_i}_{\hat{=}\beta_i \text{ random intercept captures time-constant confounder}} + \beta_2 x_{ij} \quad \Rightarrow E(Y_{ij} - Y_{ik}) = \beta_1 (x_{ij} - x_{ik})$$

Challenges

- data is correlated, but decreasing correlation over time
- drop-outs / missing data
- time-varying covariates:
  - determine lag structure
  - covariate endogeneity: response influences later covariate value

Problems of simple methods:

- linear regression: - residuals are not independent
- cross-sectional (looking at each time point individually):
  - only population trend, same trend assumed for everyone
  - assumes measures at same time points
- linear regression for each individual: - very little data for each model

Marginal Model: - model marginal correlation and account for it with robust standard errors

↳ Model $E(Y_{ij}) = \beta_0 + \beta_1 t_{ij}$ and a model for $Var(Y_{ij})$ & $Corr(Y_{ij}, Y_{ik})$

Mixed Models: - model with subject specific means

=> conditional models

↳ parameter estimation with marginal likelihood where random effects are integrated out

- A mixed model implies a specific marginal model, but not the other way around

2) Graphical display of longitudinal data

- show original data
- make trends visible
- highlight cross-sectional and longitudinal patterns

Different plot types

- Spaghetti plot
- boxplott for each time point (cross-sectional analysis)
- Lasagna plot (colums: time; row: subject)
  ↳ ordering controlls visual impression

Smooth means

- for irregular data (measures at different time points)
- smoothing methods assume i.i.d. ε_{ij}

Exploring Correlation structure

- Correlation of measurements shrink over time
  ↳ look at correlation w.r.t. time difference

3) Linear mixed model

3.1) Motivation: two-stage analysis

↳ 1) separate linear model for each subject: $Y_{ij} = \beta_{i0} + \beta_{i1} \cdot t_{ij} + \varepsilon_{ij}$

↳ 2) linear Model for coefficients, e.g. separated by gender: $\underset{\substack{\uparrow \\ \text{intercept difference} \\ \text{of genders}}}{\beta_{i0} = \beta_0 + \beta_1 g_i + b_{i1}}$; $\underset{\substack{\uparrow \\ \text{slope difference} \\ \text{of genders}}}{\beta_{i1} = \beta_2 + \beta_3 g_i + b_{i2}}$

$\Rightarrow$ combining: $Y_{ij} = \left[\beta_0 + \beta_1 g_i + (\beta_2 + \beta_3 g_i) \cdot t_{ij}\right] + \left[b_{i1} + b_{i2} t_{ij}\right] + \varepsilon_{ij}$

$\Rightarrow$ in general: 1) $Y_i = \underset{\substack{\uparrow \\ \text{time}}}{Z_i} \beta_i + \varepsilon_i$, 2) $\beta_i = \underset{\substack{\uparrow \\ \text{gender}}}{K_i} \beta + b_i$ $\Rightarrow Y_i = \underbrace{Z_i K_i}_{=: X_i} \beta + Z_i b_i + \varepsilon_i$

- bad for small number of obs per subject
- information loss because of 2-stage modelling
- in stage 2 not β_i but $\hat{\beta}_i$ is estimated $\Rightarrow$ more uncertainty

$\Rightarrow$ better combine both in one model

3.2) Longitudinal Linear Mixed Model

$$Y_i = X_i \beta + Z_i b_i + \varepsilon_i \quad \begin{cases} b_i \sim N_q(0, D) \\ \varepsilon_i \sim N_{n_i}(0, \Sigma_i) \end{cases} \quad b_i \;\&\; \varepsilon_i \;\; \forall i \text{ independent}$$

- X_i has time-constant & time-varying covariates (for example also time)
- $b_1, \dots, b_N$ independent, but $b_{i1}, \dots, b_{iq}$ not independent (covariance D)
- special case of general linear mixed model
  ↳ longitudinal LMM assumes Y can be devided into independent Y_i

3.3) Marginal vs. Conditional

$$Y_i = X_i\beta + Z_i b_i + \varepsilon_i \;;\; b_i \sim N_q(0, D),\; \varepsilon_i \sim N_{n_i}(0, \Sigma_i)$$

Conditional

$E(Y_i|b_i) = X_i\beta + Z_i b_i$

$Cov(Y_i|b_i) = \Sigma_i$

- random effects as subject-specific deviations from mean
- conditional independence if $\Sigma_i = \sigma_i^2 \cdot I_{n_i}$
- all b_i need to be estimated
  ↳ lots of parameters to estimate

Marginal

$E(Y_i) = X_i\beta$

$Cov(Y_i) = \underbrace{Z_i D Z_i^T}_{\text{subject's deviation from population}} + \underbrace{\Sigma_i}_{\text{deviation from subject-specific mean}}$

- random effects induce correlation structure

- conditional → marginal always possible
- marginal → conditional not possible : Marginal does not give distr. of $Y_i|b_i$
- fixed effects β have same interpretation in marginal & conditional

3.4) Interaction between time and covariate

$$E(Y_{ij}) = \underset{\uparrow\ \text{intercept}}{\beta_0} + \underset{\uparrow\ \text{population trend over time}}{\beta_1 t_{ij}} + \underset{\uparrow\ \text{gender diff. in intercept}}{\beta_2 g_i} + \underset{\uparrow\ \text{gender diff. in growth rate}}{\beta_3 g_i t_{ij}}$$

Only linear, but polynomial in time would also be possible : t_{ij}^p

Time trend with break point : piecewise linear: $E(Y_{ij}) = \beta_0 + \beta_1 t_{ij} + \beta_2 (t_{ij} - \tau_1)_+$

Time as a factor : $E(Y_{ij}) = \beta_1 \cdot 1\!\!1_{\{t_{ij}=t_1\}} + \dots + \beta_p \cdot 1\!\!1_{\{t_{ij}=t_p\}}$

3.5) Interaction with time and random intercept:

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \underset{\text{gender-specific intercept}}{\beta_2 g_i} + \underset{\text{gender specific slope}}{\beta_3 g_i t_{ij}} + \underset{\text{random intercept (subject specific)}}{b_i} + \varepsilon_{ij}$$

$b_i \sim N(0, d_{11}),\; \varepsilon_i \sim N_{n_i}(0, \sigma^2 I_{n_i})$

$\Rightarrow Cov(Y_{ij}, Y_{ik}) = d_{11} + \sigma^2 \cdot 1\!\!1_{\{j=k\}}$; $Corr(Y_{ij}, Y_{ik}) = \frac{d_{11}}{d_{11}+\sigma^2} = \rho \Rightarrow Corr(Y_i) = \begin{pmatrix} 1 & & \rho \\ & \ddots & \\ \rho & & 1 \end{pmatrix}$

↳ Correlation same for all subjects
↳ constant over time

3.6) Random Intercept & Random slope

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + b_{i1} + b_{i2} t_{ij} + \varepsilon_{ij}$$

$;\; b_i = \begin{pmatrix} b_{i1} \\ b_{i2} \end{pmatrix} \sim N_2(0, D),\; D = \begin{pmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{pmatrix}$

$Z = \begin{pmatrix} 1 & t_{i1} \\ \vdots & \vdots \\ 1 & t_{in_i} \end{pmatrix}$

R
- lme in nlme (Y~x, random = ~1|z)
- lmer in lme4 (Y~x+(1|z)) ; REML = TRUE is default
- gam/gamm in mgcv (Y~x+s(z, bs="re"))

4) Estimation

4.1) marginal model

$Y_i = X_i\beta + Z_i b_i + \varepsilon_i$; $b_i \sim N_q(0,D)$; $\varepsilon_i \sim N_{n_i}(0,\Sigma)$; b_i, ε_i $\forall i$ independent

$\Rightarrow$ marginal: $Y_i \sim N(X_i\beta, Z_i D Z_i^T + \Sigma_i)$

↳ im normalen LMM ist d_{11} eine Varianz und muss positiv sein

↳ im marginal model ist d_{11} eine Covarianz, weil $Cov(Y_{ij}, Y_{ik}) = d_{11} + \sigma^2 \mathbb{1}_{\{j=k\}}$

also muss d_{11} nicht zwingend positiv sein

Terminology: - α is vector of variance/covariance parameters for $V_i(\alpha) = Z_i D Z_i^T + \Sigma$

$\Rightarrow V(\alpha) = diag(V_1(\alpha), \dots, V_N(\alpha))$ since subjects are independent

- Θ is vector with all parameters for marginal model $\Theta = \begin{pmatrix}\beta \\ \alpha\end{pmatrix}$

- if $b_i \overset{iid}{\sim} N(0, d_{11})$ independent of $\varepsilon_{ij} \overset{iid}{\sim} N(0,\sigma^2)$ $\Rightarrow V_i = \begin{pmatrix} v^2 & \rho v^2 \\ \rho v^2 & v^2 \end{pmatrix}$; $v^2 = d_{11} + \sigma^2$, $\rho = \frac{d_{11}}{d_{11}+\sigma^2}$

4.2) Estimating fixed effects β

- Likelihood is density of multidimensional normal distribution

β estimates depend on covariance specifications

$\Rightarrow$ gives $\hat{\beta}_{ML}(\alpha) = (X^T V(\alpha)^{-1} X)^{-1} X^T V(\alpha)^{-1} Y$ (is best linear unbiased (BLUE))

↳ for know α the ML estimation of β leads to weighted Least Squares:

$(y - X\beta)^{-1} W (y - X\beta)$ with $W = V(\alpha)^{-1}$

↳ still depends on α, which needs to be estimated

4.3) Estimating α (via profile Likelihood or Restricted Likelihood)

- plugging $\hat{\beta}_{ML}(\alpha)$ in marginal model likelihood gives profile likelihood

↳ gives ML estimate $\hat{\alpha}_{ML}$

$\Rightarrow$ has downward bias for Variance, because for $\hat{\sigma}$ we first have to estimate $\hat{\mu}$ because we don't know μ

$\Rightarrow$ that's why $\hat{\sigma} = \frac{1}{N-1} \cdot \sum_{i=1}^{N} (Y_i - \hat{\mu})^2$ is used (corresponds to REML)

↳ this Idea applied to LMM is REML

4.4) Numerical Calculation

- with EM or Newton Raphson

4.5) Prediction for already known individuals

- (marginal:) $b_i \sim N_q(0,D)$; (conditional): $Y_i | b_i \sim N_{n_i}(X_i\beta + Z_i b_i, \Sigma_i)$

- Bayesian approach: - $f(b_i | Y_i = y_i) = \frac{f(y_i|b_i) \cdot f(b_i)}{\int f(y_i|b_i) \cdot f(b_i)\, db_i}$

$\Rightarrow b_i$ estimated as $\hat{b}_i(\Theta) = E(b_i | Y_i = y_i)$

↳ Unbiased, so $E(\hat{b}_i(\Theta)) = E(b_i) = 0$

- Shrinkage: - prediction of random effects $\hat{b}_i$ have less variability than if the subject effects are estimated as fixed effects

- $\hat{\beta}$ and $\hat{b}$ are solutions to Henderson Mixed Model Equations

↳ equations arise when maximizing the log-Likelihood of Y and b

$\left.\begin{array}{l}\text{- Henderson equations: conditional model perspective} \\ \text{- estimating } \beta \text{ like in 4.2: marginal model perspective}\end{array}\right\}$ give same fixed effect estimates

5) Inference with LLMM

5.1) Covariance Matrix for $\hat{\beta}$ (Variance of fixed effects)

$Y_i \sim N_{n_i}(X_i\beta, V_i(\alpha))$ => $\hat{\beta}_{ML}(\alpha) = (X^T V(\alpha)^{-1} X)^{-1} X^T V(\alpha)^{-1} Y$ with α_{ML} or α_{REML}

↳ basically weighted LS estimator $\hat{\beta}_W = (X^T W X)^{-1} X^T W Y$ with $W = V^{-1}(\alpha)$

=> $E(\hat{\beta}_W) = \beta$

$Cov(\hat{\beta}_{ML}(\alpha)) = (X^T W X)^{-1} (X^T W \, Cov(Y) W X)(X^T W X)^{-1}$ with $W = V^{-1}(\alpha)$

$\overset{W=V^{-1}}{=} (X^T V^{-1}(\alpha) X)^{-1}$ } not robust against misspecification of $V(\alpha)$

(because $Cov(Y) = V$ if correctly specified)

↳ doesn't include uncertainty from estimating α

- for robustness against misspecification of $Cov(Y_i) = V_i$ in $Cov(\hat{\beta}_{ML}(\alpha)) = (X^T V^{-1} X)(X^T V^{-1} Cov(Y) V^{-1} X)(X^T V^{-1} X)$

=> Sandwich Estimator: replace V by $V(\alpha)$ and $Cov(Y)$ by empirical covariance

↳ better for high N

↳ consistent for $N \to \infty$ if $E(Y) = X\beta$ correctly specified

5.2) Inference for fixed effects

Hypothesis Testing for β => $H_0: L\beta = 0$ vs $H_1: L\beta \neq 0$

↳ e.g. $H_0: \beta_1 = 0$ => $L = [1\ 0\ 0 \dots]$

$H_0: \beta_1 = \beta_2 = 0$ => $L = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \end{bmatrix}$

- Wald-Test:
  - Test statistic under H_0 χ^2 distributed
  - ignores uncertainty of $\hat{\alpha}$ and underestimates uncertainty of $\hat{\beta}$

  => better with t- and F-Test

- Likelihood ratio Test:
  - $H_0: \beta \in \Theta_0 \subset \Theta$ vs. $H_1: \beta \in \Theta$
  - Test statistic: $\lambda_N = -2 \log\left(\frac{L(\hat{\theta}_{ML,0})}{L(\hat{\theta}_{ML})}\right)$
  - not possible for REML, because REML transforms data

  (↳ but if we do Test for α instead of β, we should use LR-Test)

5.3) Inference for random effects α

- for example H_0: only random intercept, $d_{12} = d_{22} = 0$ vs H_1: RI & RS: $d_{12} \neq 0, d_{22} \neq 0$

=> LR-Test, also possible with REML

↳ problematic because 0 is on boundary of parameter space $[0, \infty)$

↳ assumption for Test statistic to be χ^2_2 is that it's not

=> p-Values are too high ↳ tackle with mixture distr. of χ^2_1 and χ^2_2

- anova() refits with ML to do a LR-Test

6) Extensions of LLMM

6.1) Additive Models

- penalized splines: - Approximate smooth $E(Y_i) = m(x_i) \approx \sum_{j=1}^{p} \beta_j x^j + \sum_{k=1}^{q} b_k (x - \kappa_k)_+^p$
  - $\Rightarrow$ piecewise polynomial degree p
  - estimation with penalty for b: $\min_{\beta, b} \left(\|y - X\beta - Zb\|^2 + \lambda \|b\|^2 \right)$
    - ↳ this is equivalent to minimizing penalized Log-Likelihood with $d^2 = \frac{\sigma^2}{\lambda}$
    - $\Rightarrow$ estimate λ from data using (RE)ML: $\lambda = \frac{\sigma^2}{d^2}$
      - ↳ Variance of rand-eff as regularization parameter

6.2) Serial correlation

- longitudinal Linear mixed model:

$$Y_i = X_i \beta + Z_i b_i + \varepsilon_i \quad ; \quad b_i \overset{iid}{\sim} N_q(0, D) \quad ; \quad \varepsilon_i \sim N_{n_i}(0, \Sigma_i) \quad ; \quad b_i, \varepsilon_i \;\; \forall i \text{ independent}$$

$$\Rightarrow Cov(Y_i) = Z_i D Z_i^T + \Sigma_i \quad \& \quad Cov(Y_i | b_i) = \Sigma_i = \sigma^2 \cdot I_{n_i} \quad (\text{↯ restriction})$$

- Idea: decompose $\varepsilon_i = \underbrace{\varepsilon_{(1)i}}_{\text{uncorrelated}} + \underbrace{\varepsilon_{(2)i}}_{\text{correlated}}$; $Cov(\varepsilon_{(2)i}) = \tau^2 H_i \quad \Rightarrow \Sigma_i = \tau^2 H_i + \sigma^2 I_{n_i}$

additional error

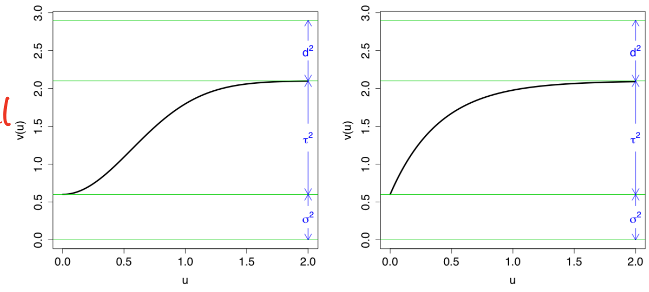
serial correlation

σ^2 is called „nugget effect"

↳ H_i is assumed to be a function of the temporal distance $h_{ijk} = g(|t_{ij} - t_{ik}|)$; $g(u) = 0,\; u \to \infty$
  - ↳ eg. $g(u) = \exp(-\phi u)$ or $g(u) = \exp(-\phi u^2)$
    - ↳ lme(): corr = corExp(form = ~1 | Subject)
  - $\Rightarrow$ only parameters τ^2 and ϕ

- Semi-Variogram
  - to assess serial correlation ; describe covariance of stationary stochastic process
  - $v(u) = \frac{1}{2} \cdot E\left[(Y(t) - Y(t-u))^2 \right]$
  - example: Random intercept model: $Y_{ij} = x_{ij}^T \beta + \overbrace{b_i + \varepsilon_{(1)ij} + \varepsilon_{(2)ij}}^{\text{Expectation } 0, \text{ Variance } d^2 + \sigma^2 + \tau^2}$

squared exponential



exponential

u is the time difference

- in practice: $v(\cdot)$ has to be estimated with residuals $r_{ij} = y_{ij} - x_{ij}^T \hat{\beta}$: $\hat{v}_{ijk} = \frac{(r_{ij} - r_{ik})^2}{2}$
- nlme::Variogram

6.3) Variance heterogeneity

- Variance structure might change over time/covariates $\Rightarrow Var(\varepsilon_{ij}) = g(x_{ij}, (\beta, b), \alpha)$
- nlme : varFixed, varIdent, varPower, ...

7) Model choice & diagnostics

- because $E(b_i)=0$, all random effect covariates should be a linear transformation of fixed effects

7.1) Model diagnostics

- when plotting residuals against covariates or estimated values, there should be no trends
  ↳ a spread that gets wider means a random slope is missing
- $Cov(Y_i - X_i\beta) = V_i$ => Residuals correlated & heteroscedastic
  => normalized residuals $r_i^* = L_i^{-1} r_i$ $(V_i = L_i^T L_i)$ are uncorrelated with variance 1
- squared Mahalanobis distance: $d_i = r_i^{*T} r_i^* = (y_i - X_i\hat{\beta})^T V_i^{-1}(\alpha)(y_i - X_i\hat{\beta})$
  ↳ measure of distance between observed and fitted

Choice of Covariance Structure

- semi-variogram: $\frac{1}{2}E\left[(r_{ij}^* - r_{ik}^*)^2\right] = \frac{1}{2}\left[Var(r_{ij}^*) + Var(r_{ik}^*) - 2Cov(r_{ij}^*, r_{ik}^*)\right] = \frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 1 + 0 = 1$
  ↳ so it should randomly fluctuate around 1

7.2) Model choice

- Compare 2 Models:
  - if same number of parameters: compare likelihood
  - if nested: use a LR test (anova)
  - use AIC/BIC $(-2\cdot\ell + 2\cdot df \;/\; -2\cdot\ell + \ln(n)\cdot df)$
    ↳ if models are nested, equivalent to LR-Test
    ↳ minimizing AIC minimizes KL-distance to true model
- Marginal AIC:
  - AIC based on marginal log likelihood: $\log f(y|\beta,\alpha)$
  - appropriate when focus on population fixed effects
  - preference for models with fewer random effects induced
  - Should not be used with REML (not compareable
- Conditional AIC:
  - based on conditional log likelihood: $\log f(y|b,\beta,\alpha)$
  - appropriate when focus on random effects
  - possible for ML and REML

7.3) Assumptions & confounding

- two assumptions for LMM: i) $E(\varepsilon_{ij}|x_{ij},b_i)=0$; ii) $E(b_i|x_{ij})=0$
- decompose $Y_{ij} = \beta_0 + \beta_B\bar{x}_i + \beta_W(x_{ij}-\bar{x}_i) + b_i + \varepsilon_{ij}$
  ↳ β_B: between subjects information ; β_W: within-subject information
  ↳ both can be confounded
- fixed vs random effect b_i
  ↳ Idea: If doubts about random effect assumptions, just model b_i as fixed effects
  ↳ less efficient if random effect assumptions satisfied
  ↳ maybe too small n_i to estimate random effects as fixed effects
  ↳ additional degrees of freedom used

8) Non-normal longitudinal data (binary / count / non-normal continuous)

8.2) GLMs
- every exponential family distribution defines relationship between expected value & variance
  - ↳ in any other case than gaussian, expected value and variance can't be modelled separately
- Assumptions:
  - Y_i has density $f(y_i | \theta_i, \phi)$ from exp fam
  - $\mu_i = E(Y_i)$ is of form $\mu_i = h(x_i^T \beta)$
- deriving the log-likelihood and setting it to 0 gives score equations for ML estimators

8.3) extending GLMs to longitudinal data
- until now, we had $E(Y_i) = E(X_i\beta + Z_i b_i) = X_i\beta + Z_i \cdot E(b_i) = X_i\beta$ in the mixed model
  and $Y_i \sim N(X_i\beta, Z_i D Z_i^T + \Sigma_i)$ in the marginal view

=> this now changes:
- mixed model focusses on subjects
- marginal model focusses on population

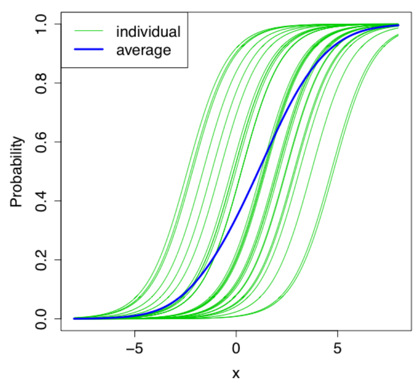
Overview marginal model:
- $Var(Y_{ij}) = \phi \cdot v(\mu_{ij})$
- $E(Y_{ij}) = h(x_{ij}^T \beta)$
- difference to GLM: $Y_{i1}, \dots, Y_{in_i}$ are conditionally (on covariates) associated

↳ Example: $Y_{ij} \in \{0,1\}$ logistic model => Marginal model:
$$\text{logit } E(Y_{ij}) = \beta_0 + \beta_1 x_{ij}$$
$$Var(Y_{ij}) = \mu_{ij}(1-\mu_{ij})$$
$$Corr(Y_{ij}, Y_{ik}) = \rho(\mu_{ij}, \mu_{ik}; \alpha)$$
↳ separate modelling of mean and correlation
↳ inference about population mean
↳ no likelihood (see GEE)

=> Mixed Model: $\text{logit } E(Y_{ij}) = (\beta_0^* + b_i) + \beta_1^* x_{ij}$; $b_i \overset{iid}{\sim} (0, d^2)$
↳ inference about individuals
↳ can model unequally spaced time points
↳ likelihood based inference

=> Transition Model: $\text{logit } E(Y_{ij} | Y_{ij-1}, \dots, Y_{i1}) = \beta_0^{**} + \beta_1^{**} x_{ij} + \gamma \cdot Y_{ij-1}$
↳ current state depends on last observation
↳ might be a meanigful way to think of underlying process

- marginal & mixed logistic model:



- β_1 controlls the slope of the function
- random intercept just shifts it left / right

=> individual effect is always bigger or equal than population effect : $|\beta_1| \leq |\beta_1^*|$

9) GLMM: Conditional model for non-linear relations (exp fam: $f = \exp\left(\phi^{-1}(y_{ij}\theta_{ij} - \psi(\theta_{ij})) + c(y_{ij}, \phi)\right)$)

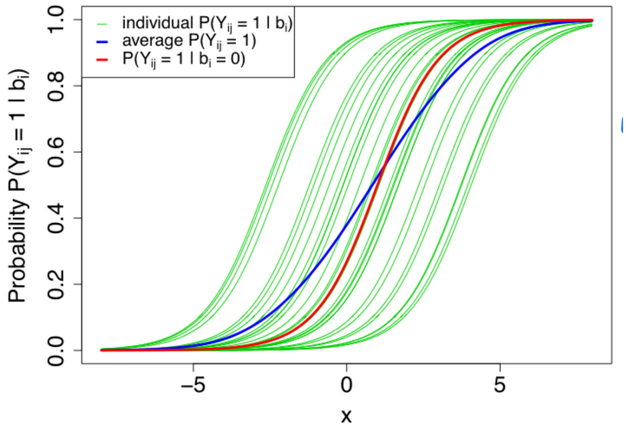
9.1) Introduction

- conditional on b_i; Y_{ij} are independent for all j and have exp fam distr. $f(y_{ij} | b_i, \beta, \phi)$
- conditional expectation $\mu_{ij} = E(Y_{ij} | b_i) = h(x_{ij}^T \beta + z_{ij}^T b_i)$
  - ↳ marginal expectation: $E(Y_{ij}) = E(E(Y_{ij} | b_i)) = E(h(x_{ij}^T \beta + z_{ij}^T b_i)) \neq h(E(x_{ij}\beta + z_{ij}^T b_i))$
- $b_i \overset{iid}{\sim} N_q(0, D)$
- distr. Assumption: $f(y | b, \beta, \phi)$
- Systematic component: $\eta_{ij} = x_{ij}^T \beta + z_{ij}^T b_i$
- Link function: $g(\mu_{ij}) = x_{ij}^T \beta + z_{ij}^T b_i$
- logistic regression: $Var(Y_{ij} | b_i) = \mu_{ij}(1 - \mu_{ij})$; $g(\mu_{ij}) = \log\left(\frac{\mu_{ij}}{1 - \mu_{ij}}\right)$
- Poisson regression: $Var(Y_{ij} | b_i) = E(Y_{ij} | b_i)$; $g(\mu_{ij}) = \log(\mu_{ij})$

can't switch between conditional & marginal (population level)

9.2) Parameter Interpretation

- always watch for link function in interpretation



9.3) Estimation & Inference

- marginal likelihood for subject i: $f(y_i | \beta, D, \phi) = \int_{\mathbb{R}^q} \prod_{j=1}^{n_i} f(y_{ij} | b_i, \beta, \phi) \cdot f(b_i | D)\, db_i$
- ⇒ Likelihood: $L(\beta, D, \phi) = \prod_{i=1}^{N} f(y_i | \beta, D, \phi)$
  - ↳ Product over high dimensional Integral ⇒ approximate data / integrand / integral

Approximate the data with Penalized Quasi Likelihood (PQL)

- use Taylor to approximate data and then apply LMM
- used in gamm() in mgcv
- only accurate for LMM

Approximate the integrand with Laplace approximation (basically 2. order Taylor of integrand)

- such that analytically solvable

Approximate the integral: Gaussian quadrature

↳ Idea: Approximate Integral with Riemann Sums

Inference

- $\hat{\beta}_{ML} \overset{a}{\sim} N(\beta, I^{-1})$ with I: Fisher information matrix
- PQL leads to estimation on pseudo data ⇒ no LR-Test

Remarks

- for binary data, only use random intercept as random effect

10) Marginal Models for non-normal responses (GEE)

10.1) Marginal Model

- expected value and association are modeled separately (unlike GLMM)
  - ↳ expected value model: $E(Y_{ij}) = h(x_{ij}^T \beta)$
  - ↳ variance depends on marginal mean: $Var(Y_{ij}) = \phi\, v(\mu_{ij})$ (= ϕ if homoscedastic)
  - ↳ correlation is function of marginal means and additional parameter: $Corr(Y_{ij}, Y_{ik}) = \rho(\mu_{ij}, \mu_{ik}, \alpha)$
- example: normally distr. response: $Y_i \sim N_{n_i}(X_i\beta, V_i)$; Link-function is identity
  - ↳ fully specified by first two moments
    - ↳ not the case for other distributions, since we additionally need association parameters (difficult)

=> Solution: GEE => we only estimate first two moments of response distribution as simplification

10.2) GEE - Principle

- focusses on modelling the mean
- Idea: Minimize $[Y - \mu(\beta)]^T V^{-1} [Y - \mu(\beta)]$ with working variance V and $\mu_{ij} = h(x_{ij}^T \beta)$ to get $\hat{\beta}_{GEE}$
  - ↳ Minimum fulfills score equation: $\sum_{i=1}^{n} \left(\frac{\delta \mu_i}{\delta \beta}\right)^T V_i^{-1} (y_i - \mu_i(\beta)) = 0$

Comparison

- GLS:

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{N} \mathbf{X}_i^T \mathbf{V}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta}) = \mathbf{0}.$$

GLS is special case of GEE with $\mu_i(\beta) = X_i\beta$

(actual score function derived from Gaussian log-likelihood)

- GLM:

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{N} \frac{\partial \theta_i}{\partial \boldsymbol{\beta}} [y_i - \psi'(\theta_i)] = \sum_{i=1}^{N} \frac{\partial \mu_i}{\partial \boldsymbol{\beta}} v_i^{-1} (y_i - \mu_i) = \mathbf{0}.$$

(actual score function derived from GLM log-likelihood)

- GEE:

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{N} \left(\frac{\partial \boldsymbol{\mu}_i}{\partial \boldsymbol{\beta}}\right)^T \mathbf{V}_i^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}.$$

(solution minimizes Mahalanobis distance between $\hat{\boldsymbol{\mu}}_i$ and $\mathbf{Y}_i$)

10.4) Properties and inference

- $\hat{\beta}_{GEE}$ is consistent for β as $N \to \infty$, even if Covariance assumption is incorrect
- $\hat{\beta}_{GEE}$ is asymptotically multivariate normal distributed

- Advantages of GEE:
  - almost as precise as ML estimator
  - for multivariate normal similar to GLS estimator
  - robust against misspecification of the variance structure
- Disadvantages:
  - more efficient if Covariance correctly specified
  - robustness is asymptotic for $N \to \infty$
  - we don't get a completely specified distribution out

10.6) GLMM & GEE

- GEE models expected value and covariance separately
- GEE allows interpretation on population level
- GLMM assumes measurements are independent conditional on random effects
- GLMM: Inference on individual level

- Marginal interpretation can be derived from conditional model, but not vice versa
- GEE has no likelihood => harder model selection / comparison

11) Missing values

- Notation: - $R_{ij} = 1$ if Y_{ij} is measured, else $R_{ij} = 0$
  ↳ this devides Y_i into observed Y_i^o and missing Y_i^m
  - Completer: subject i has no missings
  - Dropout: $R_{ij} = 1$ for $j < k$ an $R_{ij} = 0$ for $j \geq k$
  - Intermitted missing: missing somewhere inbetween

11.2) 3 Mechanisms

Missing Completely at Randon (MCAR)

- Probability of missinqness is not related to responses or even Covariates

$$P(R_{ij}=1 \mid Y_i^o, Y_i^m, X_i) = P(R_{ij}=1)$$

Missing at Random

- Probability of missing is not related to the value that would have been observed

$$P(R_{ij}=1 \mid Y_i^o, Y_i^m, X_i) = P(R_{ij}=1 \mid Y_i^o, X_i)$$

↳ can be imputed as they only depend on observed data

Not missing at Random (NMAR)

- Probability of missing depends on obsered and unobserved values
- example: rich people answer less likely questions about income

11.3) dropout

- often unknown if MCAR or MAR, because we don't see subjects again

11.4) ML methods and GEE for missing values

Likelihood-based inference

- most important to distinguish (MCAR/MAR) and (NMAR)

$\Rightarrow$ density of (Y^o, Y^m, R): $f(y^o, y^m, r \mid X) = f(y^o, y^m \mid X) \cdot f(r \mid y^o, y^m, X)$

↳ for MCAR/MAR y^m can be ignored $\Rightarrow$ „ignoreable missingness"

GEE

- used for consistency under misspecified covariance
- but only consistent for MCAR
- for MAR, we can use a GEE with additional weights $\frac{1}{P(R_{ij}=1)}$

11.5) Data Analysis methods

Complete case analysis: - Only look at complete cases
- induces bias for MAR and NMAR

Available data analysis: - only use for MAR with likelihood-based methods

Impatation methods: a) last value carried forward with observed time trend

b) impute y_i^m with $f(y_i^m \mid y_i^o, X_i)$

↳ Propensity based impatation: - missing values imputed based on subjects with similar dropout probability

↳ Predictive mean matching: - predict observation k based on other subjects where observation k is not missing

c) Multiple impatation: impute multiple values and compare estimated $\hat{\beta}$

Notes from Exercises

- $Cov(\vec{v}) = E(\vec{v}\,\vec{v}^T)$ if $E(\vec{v}) = \vec{0}$
- $Cov(A\vec{v}) = A\, Cov(\vec{v})\, A^T$

- Selection models are based on the factorization

$$f(\boldsymbol{y}_i, \boldsymbol{r}_i | \boldsymbol{X}_i, \boldsymbol{\theta}, \boldsymbol{\psi}) = f(\boldsymbol{y}_i | \boldsymbol{X}_i, \boldsymbol{\theta}) f(\boldsymbol{r}_i | \boldsymbol{y}_i, \boldsymbol{X}_i, \boldsymbol{\psi})$$

- Pattern mixture models are based on the factorization

$$f(\boldsymbol{y}_i, \boldsymbol{r}_i | \boldsymbol{X}_i, \boldsymbol{\theta}, \boldsymbol{\psi}) = f(\boldsymbol{r}_i | \boldsymbol{X}_i, \boldsymbol{\psi}) f(\boldsymbol{y}_i | \boldsymbol{r}_i, \boldsymbol{X}_i, \boldsymbol{\theta}), \text{ i.e.,}$$

r_i is missingness indicator

---

marginal correlation RI-Model

↳ btw. measurements of same subject

$$
\begin{aligned}
Cov(\boldsymbol{Y}_i) &= Cov(\boldsymbol{X}_i\boldsymbol{\beta} + \boldsymbol{Z}_i\boldsymbol{b}_i + \varepsilon_i) \\
&= \underbrace{Cov(\boldsymbol{X}_i\boldsymbol{\beta})}_{=0} + Cov(\underbrace{\boldsymbol{Z}_i}_{=\mathbb{1}_{ni}} b_i) + Cov(\varepsilon_i) \\
&= Cov(\mathbb{1}_{ni} b_i) + Cov(\varepsilon_i) \\
&= d_{11} \underbrace{\mathbb{1}_{ni}\mathbb{1}_{ni}^\top}_{\text{matrix of all 1s}} + \sigma^2 \boldsymbol{I}_{n_i} \\
&= \begin{pmatrix} d_{11}+\sigma^2 & d_{11} & \cdots & d_{11} \\ d_{11} & \ddots & & \vdots \\ \vdots & & \ddots & d_{11} \\ d_{11} & \cdots & d_{11} & d_{11}+\sigma^2 \end{pmatrix}
\end{aligned}
$$

$$\Rightarrow Corr(Y_{ij}, Y_{ik}) = \frac{Cov(Y_{ij}, Y_{ik})}{\sqrt{Var(Y_{ij}) \cdot Var(Y_{ik})}} = \frac{d_{11}}{d_{11}+\sigma^2}$$

- the conditional correlation has b_i given, so $Cov(Z_i b_i) = 0$

ML

i) find ML estimator for β in dependence of α

$\Rightarrow$ gives GLS: $\hat{\beta} = (X^T V^{-1} X)^{-1} X^T V^{-1} y$

ii) find $\hat{\alpha}_{ML}$ by maximizing profile log-likelihood $\ell(\hat{\beta}(\alpha), \alpha)$

iii) set $\hat{\beta}_{ML} = \hat{\beta}(\alpha_{ML})$

REML

i) same: $\hat{\beta} = (X^T V^{-1} X)^{-1} X^T V^{-1} y$

ii) mit REML sollen die Daten für eine Schätzung von α benutzt werden, die keine Information über β enthalten, also müssen globale Trends eliminiert werden. Wir suchen also erstmal ein $A \in (n-p) \times n$ sodass $AX = 0$ $\Rightarrow E(AY) = AX\beta = 0$

$\Rightarrow AY \sim N(0, AVA^T)$ eliminiert globale trends β

$\Rightarrow$ Likelihood von AY maximieren schätzt α ohne Verzerrung von β

iii) set $\hat{\beta}_{REML} = \hat{\beta}(\alpha_{REML})$

function gls() => Marginal model with unstructured correlation structure

- Model assumption: - $Y_{ij} = X_i\beta + \delta_i$, $\delta_i \sim N(0, V_i)$

- unstructured assumption: $V_i = \sigma^2 \cdot \begin{bmatrix} 1 & & & \\ \rho_{12} & 1 & & \\ \rho_{13} & \rho_{23} & 1 & \\ \rho_{14} & \rho_{24} & \rho_{34} & 1 \end{bmatrix}$ (4 observations per subject)

this is given in R output

- gls() modelliert ein marginal model, während lme() ein mixed model, also ein conditional model, modelliert

↳ in marginal models, negative correlations are possible, in mixed models not

function gam(): - s(time, bs="ps", ...) modelliert penalized basis splines

↳ edf = 1 heißt linear

- s(subject, bs="re") modelliert random intercept

=> man modelliert dann sowas wie

$$Y_{ij} = \beta_0 + f(time_{ij}) + b_{i0} + \varepsilon_{ij}$$

↳ $b_{i0} \sim N(0, d_{11})$, $\varepsilon_{ij} \sim N(0, \sigma^2)$

↳ $f(x) = \sum_{j=1}^{p} \beta_j x^j + \sum_{k=1}^{q} b_k (x - \kappa_k)_+^p$ bei piecewise polynomials

$f(x) = \sum_k B_k(time)\, \beta_k$ bei basis splines

- GLMM for binary data: - $Y_{ij} | b_i \sim Bernoulli(\mu_{ij})$ => $E(Y_{ij}|b_i) = \mu_{ij}$

- $b_i \overset{iid}{\sim} N(0, d_{11})$

- Y_i are independent given b_i

- logit Link: $\log\left(\frac{\mu_{ij}}{1-\mu_{ij}}\right) = \eta_{ij}$

- systematic: $\eta_{ij} = \beta_0 + \beta_1 \cdot time_{ij} + \beta_2 \cdot sex_i + b_{i0}$

GEE: $Y_{ij} \sim Bernoulli(\mu_{ij})$

with working correlation: $Corr(Y_{ij}, Y_{ik}) \overset{AR(1)}{=} \alpha^{|j-k|}$

↳ focus on estimating the mean while taking the working variance into account